

Repository Report — hcs-web

Heron Constructive Solutions LTD marketing website

Node.js · Express 4 · EJS · Tailwind CSS v4 | Branch `main` · 210 commits · 1 uncommitted change | 28 July 2026

Summary. The architecture is sound — clean route → controller → service → view layering, a thorough SEO meta partial, and a service layer designed so data can later move to a CMS without touching views. What is missing is the last mile. The contact form, the site's entire commercial purpose, is broken end-to-end; most page content is still unmodified Tailwind UI e-commerce demo material; and several published details (Cheshire office address and phone) appear to be placeholder data. This is a working scaffold, not a shippable site.

1. Architecture SOLID

Conventional MVC-ish layering, small and consistent throughout:

Layer	Contents	LOC
<code>routes/</code>	4 routers, all mounted at <code>/</code>	68
<code>controllers/</code>	page, content, blog, service	208
<code>services/</code>	view-model builders, in-memory data	381
<code>middleware/</code>	404 + error handler	51
<code>views/</code>	17 EJS templates	~2,100
<code>public/css/styles.css</code>	generated Tailwind output (committed)	4,038

`app.js` and `server.js` are correctly split (create-app vs. listen), which keeps the app testable. Controllers are thin and consistently use `try/catch → next(err)`. `services/pageService.js:18` `getBaseViewModel()` is a good single source of truth for site-wide data, spread into every page view-model.

Routes served: `/`, `/about`, `/studies`, `/contact` (GET + POST), `/services`, `/privacy`, `/fencing`, `/blog`, `/blog/:slug`, plus `/version`, `/sitemap.xml`, `/robots.txt`.

2. The contact form does not work CRITICAL

This is the site's primary business function — lead generation — and it is broken end-to-end.

- `views/pages/contact.ejs:155` — `action="#"`. The form never submits anywhere. `POST /contact → pageController.contactPost` and `submitContactForm` are dead code.
- Field names do not match the backend even if it did submit: the form sends `first-name`, `last-name`, `phone-number`, `email`, `message`; the handler reads `name`, `email`, `message`. Validation would reject every submission on the missing `name`.
- The view never renders `error`, `sent`, or `form.*`. The controller carefully passes all three back (`controllers/pageController.js:70-90`) and the template ignores them — users get no success or error feedback.
- Even on success, nothing happens: `submitContactForm` validates and returns. No email, no storage. Leads would be silently dropped.

3. Content status — largely unfinished scaffolding CRITICAL

The site is still substantially Tailwind UI demo content. Referenced images include:

- `tailwindcss.com/plus-assets/img/ecommerce-images/home-page-02-product-0{1..4}.jpg` — **ecommerce product photos** on a construction site
- `savvyca`, `transistor`, `tuple`, `reform`, `statamic` logos presented as clients / trust signals
- 11 `images.unsplash.com` stock avatars used as testimonial photos
- 5 `placeholder.co` grey boxes

`agent.md` already documents this honestly — "Shop by Category — NOT aligned with construction", "FAQs — Switzerland flag joke, pterodactyl", "Trusted By — Not actual clients". Real project photography is in the repo: `public/images/` holds 12 genuine fence, gate and paving shots plus real accreditation logos (CHAS, CITB, Constructionline, Safe Contractor, Living Wage) and real client logos (Knowsley Council, Plus Dane) — but most are unused by any template.

Missing brand assets. `public/images/` contains no `favicon.ico`, `favicon-32x32.png`, `favicon-16x16.png`, `apple-touch-icon.png` or `og-image.png`, yet `views/partials/head.ejs:69-73` and `site.ogImage` reference all five. Every page currently emits five broken asset links, including a broken Open Graph image on every social share.

One further broken image: the testimonial avatar URL is malformed — the `ixlib` version and the `ixid` key were both lost in an edit, so it will not render.

```
https://images.unsplash.com/photo-1550525811-e5869dd03032?ixlib=rb-=eyJhcnBfawQi0jEyMDd9&auto=form
at&...
^^^^^^^^^^^^^ mangled - every other URL in
the repo
has the intact "ixlib=rb-1.
2.1&ixid=" form
```

4. Data inconsistencies HIGH

The Cheshire office data conflicts between two places:

	<code>services/pageService.js:51</code>	<code>Uncommitted contact.ejs</code>
Address	456 Kingfisher Lane, Chester Business Park, Chester CH4 9QH	Windmill Lane, Preston Brook, Warrington WA4 4AZ
Phone	01298 333 212	01928 333 212

"456 Kingfisher Lane" reads as placeholder data, and `01298` is the Buxton dialling code while `01928` is Runcorn / Warrington — the service file's version looks wrong on both counts. Publishing an incorrect phone number on a lead-generation site is a real cost.

Other data bugs:

- `services/pageService.js:38` defines `facebookHHandle` (double H); `views/partials/footer.ejs:15` reads `site.facebookHandle` → always empty. All three social links render as empty `href`s regardless.
- `views/partials/head.ejs:58` sets `twitter:site` to `site.url` (a URL) instead of the handle.

- Footer links for Fences, Line Marking, Paving, Tarmac, Socials, Terms, Privacy and License are all `href="#"` — including Privacy, even though `/privacy` exists and renders.

5. The uncommitted change ATTENTION

`views/pages/contact.ejs` has +15 lines of **raw unwrapped text** appended after the closing `</div>` — office addresses, phone numbers and an SLA note with no HTML tags, no styling and no trailing newline. It will render as unstyled body text dumped at the bottom of the page, outside the layout's content container. It reads like pasted copy awaiting markup rather than a finished edit.

6. Security THIN BASELINE

Nothing is exploitable today, but the baseline is thin for a public site.

- **No helmet, no CSP, no security headers.** `agent.md` lists CSP as the number-one active to-do; it is unstarted. Four inline `<script>` blocks (`views/home.ejs:32`, `views/pages/home.ejs:44` and `:197`, `views/pages/content/fencing.ejs:158`) are the blocker, and `public/js/` does not exist yet — though `src/css/main.css:15` already declares `@source "../../public/js/**/*.js"` in anticipation.
- **No CSRF protection and no rate limiting** on `POST /contact`. Once the form is wired up, it becomes an open spam relay.
- `views/pages/blog/post.ejs:21` renders `<%= post.content %>` unescaped. Safe now (one hard-coded post); an XSS hole the moment posts come from a DB or CMS — which `services/blogService.js:5` explicitly plans for.
- Express is pinned `^4.19.2`. Current 4.x is 4.21.x, with fixes in the dependency chain since 4.19. Worth an `npm audit` and a bump.
- **Positive:** `middleware/errorHandler.js` does not leak stack traces to the client, and `.env` is gitignored and absent from the repo.

7. Build & repo hygiene

- **The README is wrong about how to run the app.** It says `npm start` twice; `package.json` has no `start` script. The only working commands are `npm run dev` and `npm run css:build`.
- **`public/css/styles.css` is both gitignored and tracked.** `.gitignore` lists it twice, but the file was committed before those rules, so gitignore has no effect and the 4,038-line generated file keeps getting committed. That is why a large share of the 210 commits are "Update styles.css". Fix with `git rm --cached public/css/styles.css` — but only alongside a real build step, since nothing else generates the CSS at deploy time.
- **No tests, no linter, no CI.** The `eslint-disable-next-line` comments in `server.js` and `middleware/errorHandler.js` suggest ESLint was intended but never configured.
- **Commit history is web-editor style.** Nearly every message is "Update <filename>", with occasional "Merge branch 'main'". No branches, no PRs, no descriptive messages — intent is hard to reconstruct from history.
- Two PDFs are committed as binaries (`HCS-Website-Design-Brief.pdf`, `hcs-web-git-report.pdf`).

8. Dead and orphaned code

- **`views/home.ejs` (459 lines) is orphaned.** The controller renders `pages/home`, so this larger, separately maintained copy at the views root is never served. It has its own commit history through last month — easy

to edit the wrong file.

- `views/pages/services` by `category/browse.ejs` — no route renders it. The directory name also contains spaces.
- `tailwind/marketing/sections/` — two raw Tailwind UI HTML files, scratch material.
- `POST /contact` handler and `submitContactForm` (see section 2).

9. Smaller items

- `BASE_URL` is redeclared identically in four services — belongs in one config module.
- The sitemap (`services/pageService.js:167`) omits `/fencing` and `/privacy` , both live routes. The comment above `PAGES` asks contributors to keep it updated; it has already drifted.
- The nav (`views/partials/header.ejs`) has four links and omits `/studies` and `/blog` , both of which exist and are in the sitemap.
- `/fencing` sets `activeNav: '/services'` , so it highlights Services — reasonable, but nothing links to `/fencing` at all.
- `views/partials/header.ejs:11` has `class="relative bg-white fixed"` — conflicting Tailwind position utilities.
- The logo loads from the absolute URL `https://heroncs.co.uk/images/HCS-Logo-v5.png` even though the file is local at `public/images/HCS-Logo-v5.png` . It will not render offline or in local dev without network.
- Several templates use relative `src="images/..."` (no leading slash) — those break on nested routes such as `/blog/:slug` .
- Both `agent.md` and `.github/copilot-instructions.md` tell agents to edit `routes/index.js` , which does not exist (it is `routes/pageRoutes.js`).
- Environment: Node v24.15.0, npm 11.12.1, `node_modules` 37 MB. Node 24 with Express 4 is fine, though it is another nudge toward bumping Express off 4.19.2.

10. Assessment

The skeleton is genuinely good — the layering is clean, the SEO/meta partial is thorough, and the service layer is set up so data can move to a CMS without touching views. What is missing is the last mile: this is a scaffold with demo content, not a shippable site.

Priority order:

1. Wire up the contact form — action, field names, feedback rendering, actual delivery. It is the site's entire purpose.
2. Fix the Cheshire office address and phone, then land the pending `contact.ejs` edit as proper markup.
3. Add the missing favicons and og-image.
4. Replace e-commerce, Unsplash and placeholder assets with the real photography already in the repo.
5. Delete or merge orphaned `views/home.ejs` before someone edits the wrong file.
6. Add `helmet` + CSP, extract inline scripts, add rate limiting to the contact POST.
7. Fix `npm start` in the README, untrack the generated CSS, add a lint/build step.

Report generated 28 July 2026 from branch `main` at commit `0075fb7`, plus one uncommitted modification to `views/pages/contact.ejs`.